Common scalar repair: exact census and rational successor

Producer research report; unsigned evidence

8 November 2026

The common scalar eliminates the finite mode proliferation caused by the entry-dependent normalizer. At equal precision it has fewer admissible tuples in every one of the 2,616 registered cells for each representation. This is component evidence, not measured execution speed. All findings here have `[pcc:unverified]` standing; no independent signature is supplied.

Let $L = 2^a$, $S = \{-L, \dots, L-1\}$, $C(T) = \prod_{s\in S}(T-s)$, $B_d(T) = C(T)/((T-d)C'(0))$, interpreted as its punctured polynomial, and $\rho_d = B_d(d)$. Compare B_d, B_d^2/ρ_d, and

$$R(T)B_d(T)^2, \qquad R(T) = \sum_{s\in S} B_s(T)/\rho_s^2.$$

At each source node, $R(s) = 1/\rho_s$, so all original node values survive. `CommonScalar.node_preserved` proves this for the actual kernel.

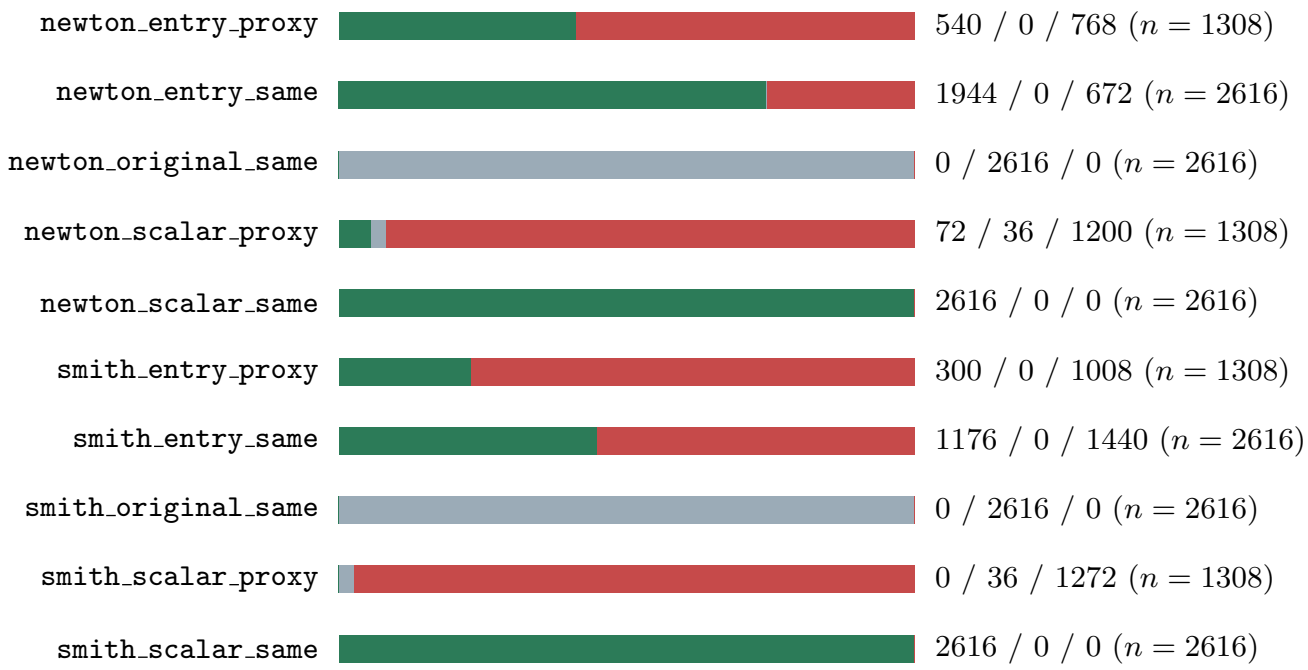
Frozen experiment. $L = 16, 32, 64$; every nonzero $t \in [-L+1, L-1]$; $\zeta = -1, 1, 3$; $T = t+2L\zeta$; $q = 16, 24, 32, 48$; Newton and Smith factorizations. All tiles and crossing rows are included. Two physical executions each emit 20,928 records. The separate grouped-generating-function analyzer checks all registered keys and recounts 41,856 integer tuple totals. Outputs are byte identical. All rational Newton and modular Smith reconstructions and the embedded controls pass (producer assertions).

Representation	Kernel	Budget	Fewer	Equal	More
Newton	entry	same	1944	0	672
Newton	scalar	same	2616	0	0
Smith	entry	same	1176	0	1440
Smith	scalar	same	2616	0	0
Newton	entry	proxy	540	0	768
Newton	scalar	proxy	72	36	1200
Smith	entry	proxy	300	0	1008
Smith	scalar	proxy	0	36	1272

Same-precision rows have $n = 2616$ per group. Proxy rows have $n = 1308$; entry precision is $\lceil 4q/3 \rceil$, scalar precision $2q$, for base $q = 16, 24$. These are conservative proxies, not a necessary actual recovery cost. No population inference, interval, p-value or timed metric is used.

Execution. One logical CPU, 2 GiB address-space cap, 900 s timeout per process; no retries or restores. Requested 2, attempted 2, completed 2, excluded 0. Both receipts record exit 0. A Lean replay also used the host, so no idle-host or valid-clock claim is made. Exact values do not depend on CPU contention.

Every outcome retained



Green: fewer candidate tuples; gray: equal; red: more. Counts are exact, recorded values from `reports/summary.json`; each bar is normalized by its printed n. The original rows are embedded A/A controls. The complete level-by-level census is `reports/per-level.csv`; no successful level was selected. Study `prime-common-scalar-20261108`; unsigned producer evidence, claim `PCC-20261108-4f9871`.

What the origin constraint permits

Zero is itself a source node. Preserving all node values therefore retains the identity at zero. The actual matrix theorem is `CommonScalar.matrix_origin`. A value-preserving jet correction $W_{\mathrm{new}} = W + CH$ satisfies $W_{\mathrm{new}}(0) = W(0)$ when $C(0) = 0$. Jet completion may change derivatives, but cannot replace this origin value.

Why the proxy loss is not a rejection

The old sufficient odd-node recovery estimate is conservative. The existing all-node recovery review supports doubled degree in its stated $u \geq 3$ regime. The first three levels already give

$$hL\left(u + \frac{u-1}{2} + \frac{u-2}{4}\right) > 4ML \quad (h > 0,\ u \geq 3,\ hu \geq 3M).$$

`RationalScalar.three_level_margin` proves this numeric inequality. The finite-precision recovery hypotheses still have to be connected to the new evaluator. This lemma alone is not that recovery theorem.

Selected successor

Replace the polynomial normalizer by one common rational denominator:

$$D(T) = \sum_s B_s(T) = \frac{C'(T)}{C'(0)}, \qquad U_d(T) = \frac{B_d(T)^2}{D(T)}.$$

At a node, $D(s) = \rho_s$ is a nonzero two-adic unit and $U_d(s) = B_d(s)$. The node statements have actual-kernel Lean proofs. For a finite homogeneous contraction with exactly M factors per term and $D \neq 0$,

$$D^M \sum_j w_j \prod_k U_{j,k} = \sum_j w_j \prod_k B_{j,k}^2.$$

`RationalScalar.clear_homogeneous_sum` proves the general identity. The cleared numerator has degree at most $M(4L-2)$, rather than $M(6L-3)$ for the polynomial scalar. The formal degree lemma is generic; its actual puncture-polynomial bridge remains open. This rational candidate has not been run as an experimental arm.

Next proof and construction costs

At chart $t + \alpha z$, $\alpha = 2L$, every $\alpha/(t-u)$, $u \neq t$, is even and two-adically integral. The explicit formulas are

$$B_t(t+\alpha z) = \rho_t \prod_{u \neq t} (1 + \alpha z/(t-u)),$$

$$B_s(t+\alpha z) = \rho_t \frac{\alpha z}{t-s} \prod_{u \neq t,s} (1 + \alpha z/(t-u)) \quad (s \neq t).$$

Thus D has unit constant coefficient and even higher coefficients; its reciprocal formal series is integral. This is the written coefficientwise argument; it is not yet a completed Lean series theorem. The pointwise `kernel_valuation` theorem explicitly assumes a unit denominator.

The proof programme is: formalize this actual local-ring bridge; connect homogeneous clearing and the source all-level finite-precision recovery; prove uniform all-level mode bounds; then charge the full recursive evaluator. A point evaluation of D by product/derivative accumulation costs $O(L)$ ring operations, not a free scalar oracle. Truncated series, inverses, basis construction, child entry queries, bit arithmetic and recovery all need charges. A constant valuation shift alone does not establish a changed asymptotic exponent. See `PLAN.md` for milestones and rejection tests.

Provenance and retained errors

Native source `555cd9430` and frozen protocol `12e310bda` were published before both execution starts. Protocol hash:

```
dedbb0086d84e5ecffba17a424a1fcfe6
b89bb3d5db0ea40a5421afc36e8d548
```

The final export has two physical rows, no valid clocks and verification `no_claim`. Canonical schemas plus explicit study-scoped native bindings pass; this checker is not claimed equivalent to the general canonical checker. Initial errors survive: unfrozen one-arm schema draft; an analyzer call made before replay completion; export missing file byte lengths. Only metadata and completed-data analysis were repaired after data; no algorithm, input, hypothesis or experimental panel was changed or rerun. Saved outputs are losslessly archived as XZ. The original adapter is in Git.

Source-linked artifacts

- `protocol.json`, `publication-receipt.json`, `source-manifest.sha256`, `export.json`, `host.txt`.
- `runs/0/panel.csv.xz`, `runs/1/panel.csv.xz`; receipts beside them.
- `reports/summary.json`, `reports/per-level.csv`, `analysis.log`, `contract-validation.log`.
- `lean/PROOF_MAP.md`, `lean/VERIFICATION.md`; all failed attempts retained.
- Prior `prime_square_kernel_20261106/RESULTS_local_20261107.md`; `prime_common_kernel_20261030/sources.md` primary-source pins; `prime_tensor_variants_20261030/reviews/matched-recovery-independent.md`.

KB unavailable (wrapper status 127); repository/atlas fallback. No priority or literature-exhaustion claim. No speedup or leaderboard promotion; independent review and trusted PCC signatures remain external obligations.